

Validation	
Overview	
• test set methods • leave-one-out cross-validation • k-fold cross-validation • variants and imported caveats to validation techniques	
Reading	
<i>The Elements of Statistical Learning</i> Trevor Hastie, Robert Tibshirani, Jerome Friedman Chapter 7-7.2, 7.10 https://hastie.su.domains/ElemStatLearn/	
Summary	• when building predictive models, two issues must be addressed ○ model assessment ▪ how well the model performs on new data ○ model selection ▪ choose the best model complexity • the key objective goes beyond simply fitting the training data to minimizing prediction error on future observations
Key Takeaways	• training error is not a reliable indicator of predictive performance • prediction error = (bias)² + variance + noise • increasing model complexity decreases bias but increases variance • model assessment requires test error estimation • cross-validation is the most common method for doing this

Validation

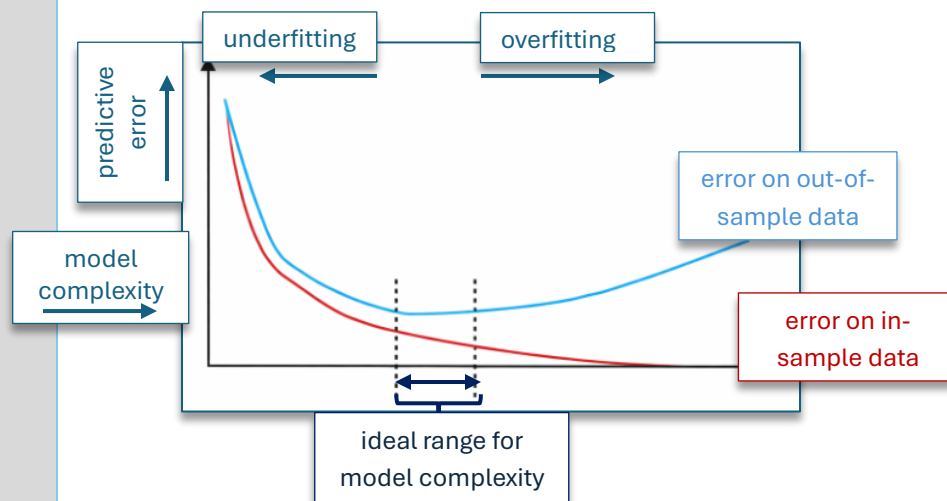
Error on Future Data

Fitting the Data

- we want the **model** to fit the **data**
 - without **over-fitting** or **under-fitting**
 - **without** becoming **over-reliant** on **training** data
- even if training **error** is **low** future **errors** can be **high**

Model Complexity and Choices

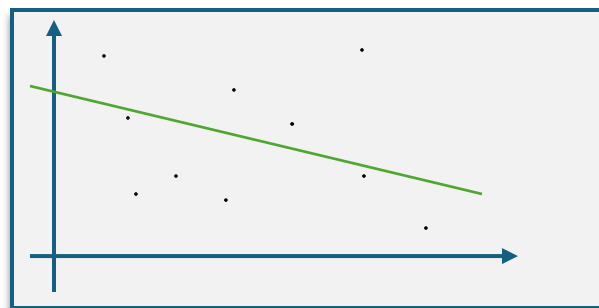
Ideal Model Complexity



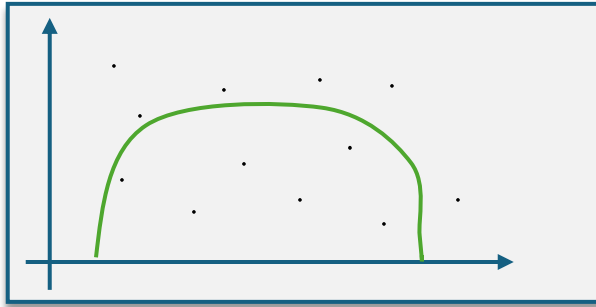
Model Complexity and Choices

Linear Regression

- often the **trend** in the data is **not obvious**
- you have to **select** the **model** that is **best** for your data



Quadratic Regression



Piecewise Non-Parametric Regression



Test-Set Validation

Test-Set Method

- **randomly** choose **30%** of the data to be a **test set**
- the **remainder** is the **training set**
- perform **regression** on the **training set**
- **estimate your future** performance with the **test set**
- this can be **achieved** with any of the **three models**
- the **optimal** model is **not** always the model with the **lowest error**
- the optimal **model** is the model that performs **best** on the **test set** **after** being **trained** on the training set

Results

- **good news**
 - simply **choose** the model with the **best** test-set **score**
- **bad news**
 - **wastes** data
 - to get the **estimate** we **lose 30%** of the data
 - if data is **sparse**, test set can **match** or not match **randomly**

Leave-One-Out Cross-Validation

LOOCV Method

- for $k = 1$ top R
 - let (x_k, y_k) be the k^{th} record
 - temporarily **remove** (x_k, y_k) from the dataset
 - **train** the remaining $R - 1$ datapoints
 - **note your error** (x_k, y_k)
- when you have done this to **all** points, report the mean **error**
- to do this you **average the mean error** from each run
- this can be **done** with **each** of the three discussed **methods**

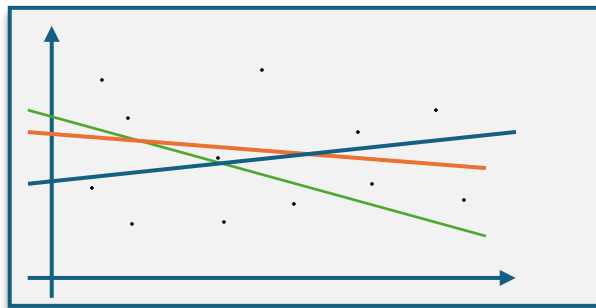
Results

- this method is **expensive**
- does **not waste** data
- this would be **unmanageable** with **large** datasets

K-Fold Cross-Validation

K-Fold Method

- randomly break the dataset into k partitions
- for the partition



- for the **red** partition
- train on **all** the **points not** in the **red** partition
- find the **test-set** sum of **errors** on the **red** points
- repeat this for the **green** and **blue** partitions
- **repeat** this **process** with **all** of the candidate **models**

Contrasting Validation Methods			
Review of Cross-Validation Methods sometimes there are algorithmic tricks for making these cheap		Downside	Upside
	Test-Set	variance unreliable estimate of future performance	cheap
	Leave-One-Out	expensive	does not waste data
	10-Fold	wastes 10% of the data 10x more expensive than test-set	only wastes 10% only 10x times more expensive instead of R times
	3-Fold	more wasteful than 10-fold more expensive than test-set	slightly better than test-set
	R-Fold	identical to leave-one-out	
Cross-Validation Applications			
Cross-Validation for Regression Hyper-Parameters	- choosing the number of hidden units in a neural net - feature selection - choosing a polynomial degree - choosing which regressor to use - parametric vs non-parametric		
Cross-Validation for Classification	- instead of computing the sum squared errors on a test set - total number of misclassifications in a test set		
CV-Based Algorithm Choice	- compute 10-fold-cv error for different model class - train the data with the algorithm that gave the best CV score		
Choosing Hyperparameters	- perform test with every combination of hyperparameters - for this hyperparameter set you will run 18 tests - hyperparameter 1 in {H11, H22} - hyperparameter 2 in {H21, H22, H23} - hyperparameter 3 in {H31, H32, H33} 2 x 3 x 3 = 18		

sometimes there are algorithmic tricks for making these cheap



Validation to Choose Bandwidth	- make a set of candidate bandwidths around the “rule of thumb” bandwidths (ex: average 5th-nearest neighbor distance) $\mathcal{H} = \{2^j h_{rot}\}_{j=-M}^M$ split the data into “train” and “validation” sets $D_{val} \cap D_{trn} = \emptyset$ choose the bandwidth $h \in \mathcal{H}$ with the best likelihood $\frac{1}{ D_{val} } \sum_{x \in D_{val}} \log(\hat{p}_{h(x)}) =$ $\left(\frac{1}{ D_{val} }\right) \sum_{x \in D_{val}} \log \left(\frac{1}{ D_{val} } \sum_{x' \in D_{val}} k_h(x', x) \right)$
Cross-Validation Alternatives	
Other Cross-Validation Issues	- can do “leave all pairs out” or “leave all n-tuples out” k-folds where each fold is independently chosen subsets
Alternatives to CV-Based Model Selection	- Model selection methods Cross-validation AIC (Akaike information criterion) BIC (Bayesian information criterion) VC dimension (Vapnik–Chervonenkis dimension)